

PrintDeck

# USER MANUAL

Navigation, setup and everyday printer control



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A short visual guide to the screens, shortcuts and product variants.

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## POCKET-SIZED PRINTER COMPANION

# Your printers. One glance away.

PrintDeck brings live printer status and camera views into one lightweight device you can keep exactly where you need it. Move between multiple printers, follow long jobs and stay informed without opening another dashboard.

**01 POCKET-SIZED**

Light and compact enough to move from the desk to the workshop.

**02 QI WIRELESS CHARGING**

Recharge on a compatible Qi pad - no cable routine required.

**03 MAGNETIC BY DESIGN**

Strong mounting for steel surfaces such as a refrigerator or printer enclosure.

**04 ADVANCED MONITORING**

Live status, controls, cameras and saved printers in one touch-first interface.

**05 CAMERA VIEWS**

Check supported local printer cameras directly from PrintDeck.

**06 LONG-RUN BATTERY**

A large battery is designed to stay with even extended print sessions.

Three purpose-built forms for PrintDeck - designed to be printed, adapted and made your own.



## 01 THE RIVERSTONE

A pocket-sized round case with MagSafe-compatible Qi charging and integrated magnets for refrigerators, tool cabinets and other metal surfaces.



AUTHOR / CREDIT

Design: kitek



## 02 THE SCUTTLE

A low, sculpted round enclosure with a clean white shell and contrasting ring and button that can be printed in your own color.



AUTHOR / CREDIT

Design: Gulios



## 03 SNAPMAKER U1 MONITOR

A miniature companion to the tool-changing Snapmaker U1. Its square PrintDeck screen becomes the dedicated status console the printer was always meant to have.



AUTHOR / CREDIT

Design: kitek

# Configuration, Updates and Factory Reset

A guided first start takes only a few minutes. Keep PrintDeck, your phone or computer and every printer on the same local Wi-Fi network.

PrintDeck Web Installer

<https://printdeck.xyz/firmware/>

## 01 OPEN INSTALLER

On a PC or Mac, open the installer in desktop Google Chrome or Microsoft Edge. Mobile browsers, Safari and Firefox cannot access the USB serial port.

## 02 CHOOSE & INSTALL

Select the latest release and the correct display. Connect a USB-C data cable, choose First installation / Factory reset, select the ESP32 serial device and wait for 100%.

## 03 JOIN PRINTDECK WI-FI

After restart, scan the QR code or join the unique open PrintDeck-XXXXXX network shown on the display. No password is required. A no-internet warning is normal during setup.

## 04 OPEN WI-FI SETUP

The Wi-Fi setup page should open after you connect. If it does not, stay connected and open <http://printdeck.local>. Use <http://192.168.4.1> only as the fallback address.

## 05 JOIN YOUR NETWORK

Choose Scan networks, select your usual Wi-Fi and enter its password. Choose Check & save Wi-Fi. PrintDeck keeps setup open and stores the details only after the network supplies an IP address.

## 06 ADD YOUR PRINTERS

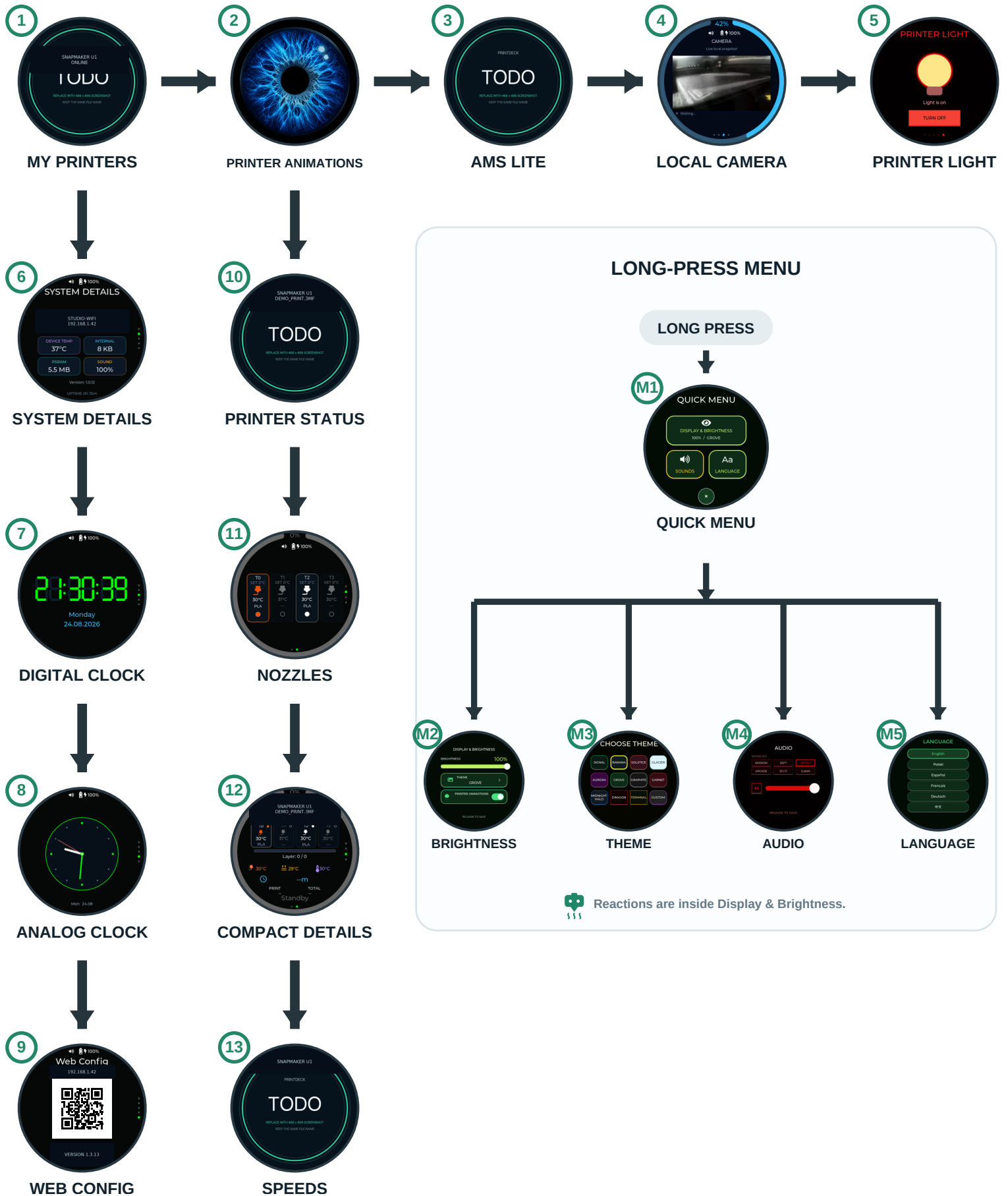
Reconnect your phone or computer to normal Wi-Fi. Open the Web Config address shown by PrintDeck, then add your Klipper / Moonraker or Bambu Lab printer.

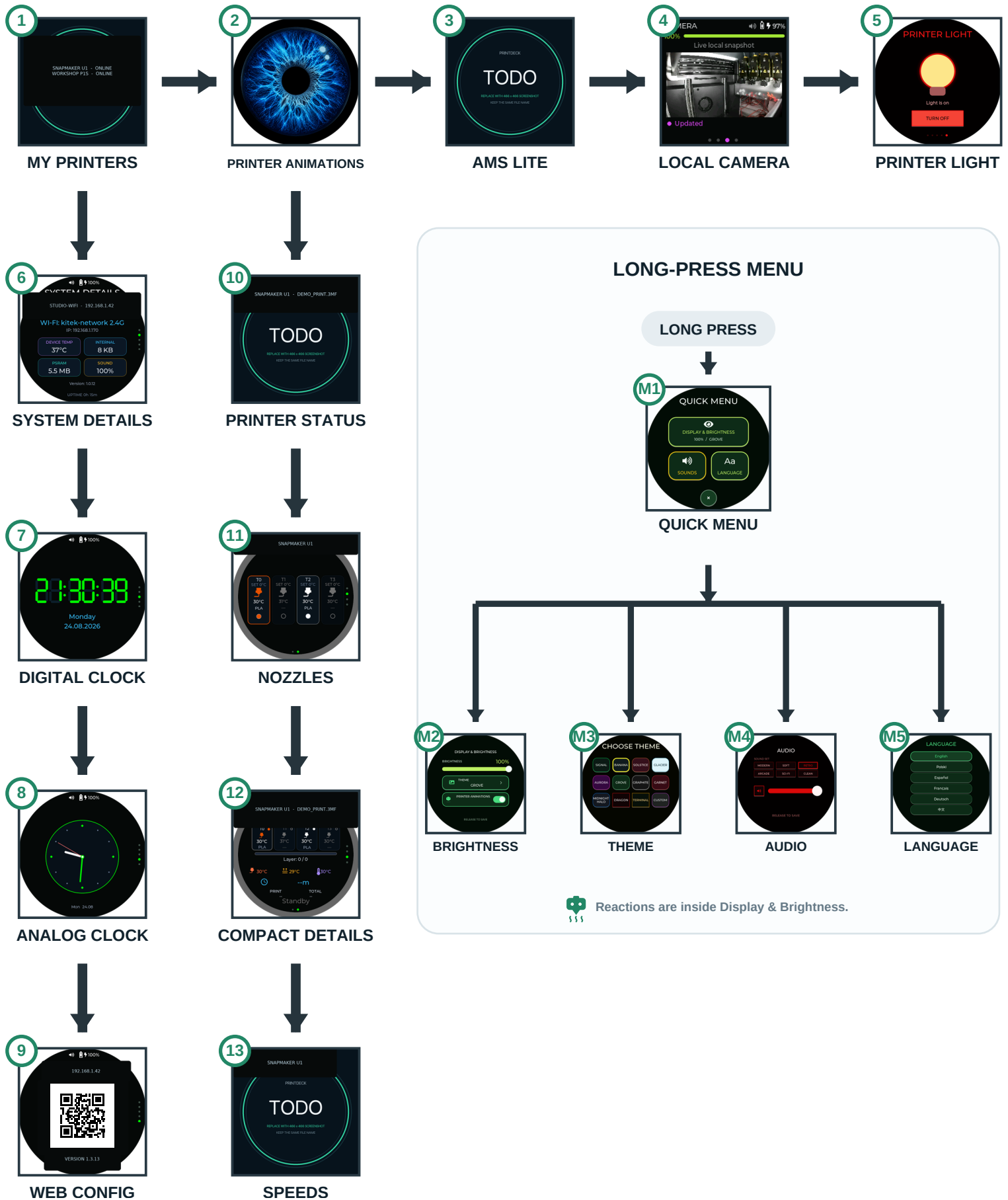
## UPDATE WITHOUT LOSING SETTINGS

For the easiest update, open Web Config and choose Device > Firmware > Check for updates, then confirm Install update. Keep PrintDeck powered while it downloads, validates and restarts. You can also return to the Web Installer, choose Firmware update and update over USB. Both update paths preserve saved Wi-Fi, printers and display settings.

## FACTORY RESET

Use the Web Installer and choose First installation / Factory reset only when setting up a new display or intentionally starting again. Confirm the full erase when prompted. This removes Wi-Fi, printers and preferences stored on PrintDeck, but it does not change anything on the printer or its controller. After restart, repeat the Wi-Fi setup shown above.





# SCREEN REFERENCE

## SCREEN NAVIGATION

- 1 MY PRINTERS**  
Your starting point for saved printers. Check availability at a glance and open a ready printer with one tap.
- 2 PRINTER ANIMATIONS**  
The first printer page when Reactions are enabled. Its animation follows heating, printing, pausing, completion and other live states.
- 3 AMS LITE**  
A clear overview of loaded materials and feeder positions whenever a compatible Bambu system is available.
- 4 LOCAL CAMERA**  
A local live view from the printer camera for a quick visual check without leaving PrintDeck.
- 5 PRINTER LIGHT**  
See and change the printer's chamber or work light from a focused control screen.
- 6 SYSTEM DETAILS**  
Review connectivity, power, memory, temperature and firmware information when checking device health.
- 7 DIGITAL CLOCK**  
A clean full-screen clock for the desk, with the current day and date always visible.

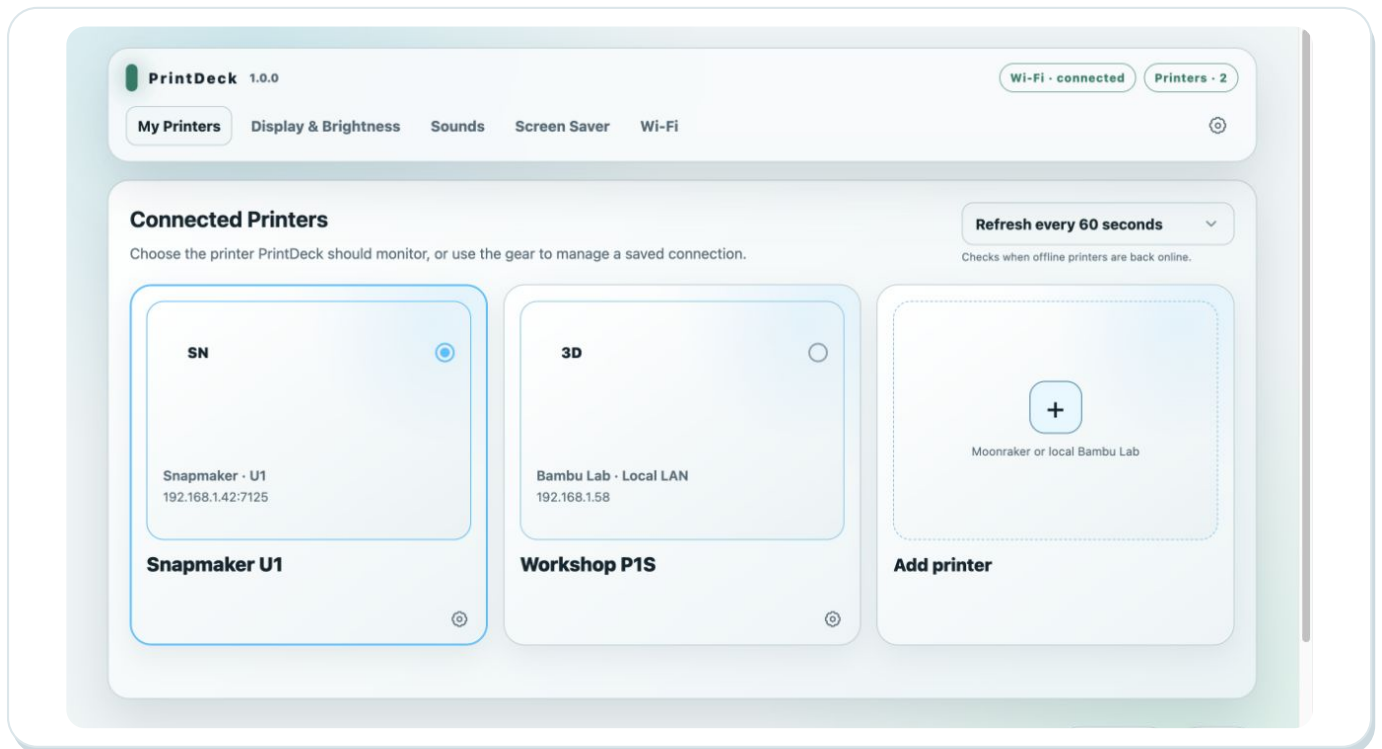
## LONG-PRESS NAVIGATION

- M1 QUICK MENU**  
Long-press any regular screen to open fast access to everyday display and sound settings.
- M3 THEME**  
Choose the visual theme that best matches your setup.
- M5 LANGUAGE**  
Choose the interface language used by the device.

- 8 ANALOG CLOCK**  
A calm analog time view that turns PrintDeck into a useful ambient desktop display.
- 9 WEB CONFIG**  
Open PrintDeck settings from a phone or computer using the shown local address or QR code.
- 10 PRINTER STATUS**  
The main live dashboard. Follow print progress, temperatures, remaining time and the current printer state.
- 11 NOZZLES**  
Monitor the active tool and available nozzles, including current and target temperatures.
- 12 COMPACT DETAILS**  
A dense but readable print summary with key timing, layer and temperature information in one place.
- 13 SPEEDS**  
Track speed, flow, fan, position and heater output while the printer is working.

# Web Configuration Panel

Configure PrintDeck from a phone, tablet or computer on the same Wi-Fi network. The panel runs locally on the device and does not require a cloud account.



## OPEN THE PANEL

### Use the address or QR code on screen (9)

Open WEB CONFIG on PrintDeck. Scan its QR code, or enter the displayed `printdeck.local` address in your browser. The IP address in System Details remains available as a fallback. Your browser device and PrintDeck must be connected to the same local Wi-Fi.

## INSTANT FEEDBACK

### Changes appear on PrintDeck

Display, theme, brightness, sound and screen-saver changes are saved to the device and reflected immediately. Keep PrintDeck in view while tuning the interface.

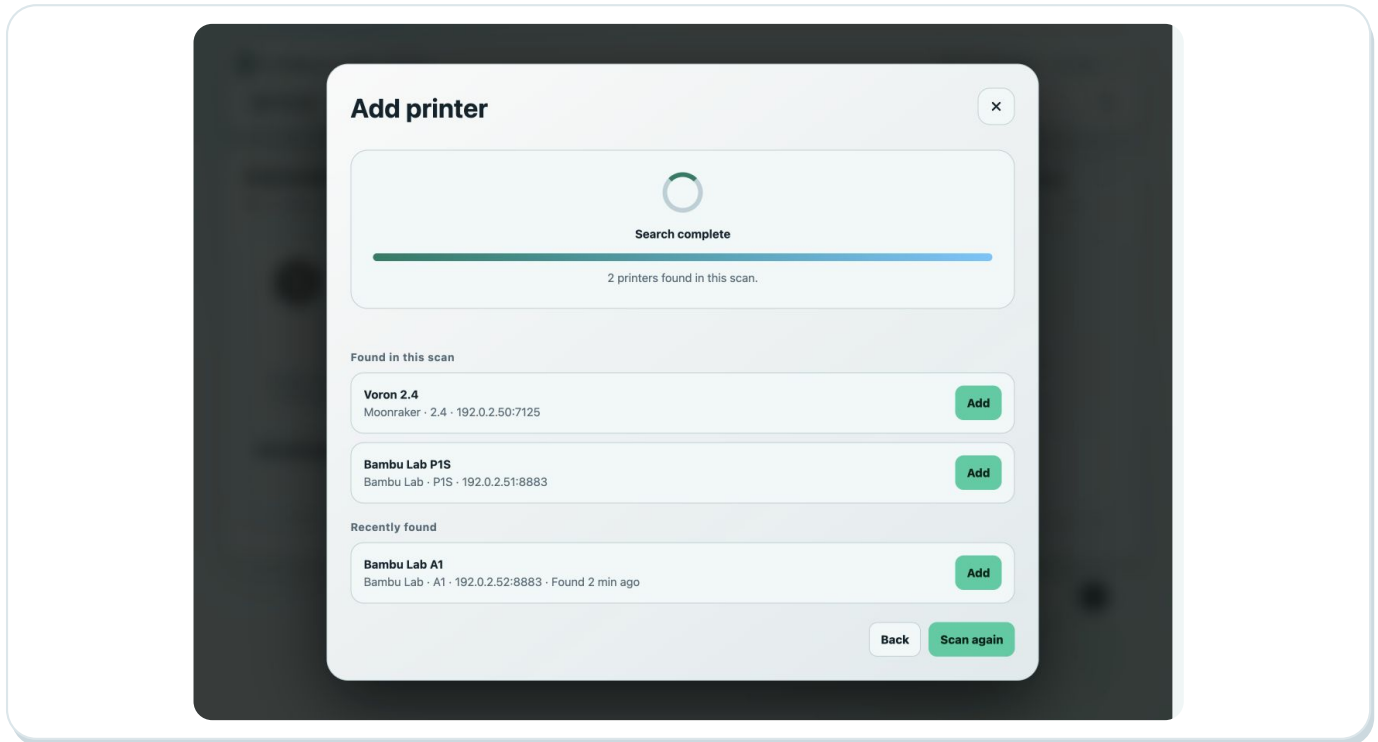
## LOCAL BY DESIGN

### Your network stays in control

The panel is served by PrintDeck on your LAN. PrintDeck does not contact external services, upload telemetry or send printer data anywhere. Bambu Lab uses local MQTT/TLS and Moonraker uses the local HTTP API, so configuration and monitoring stay inside your network.

# Web Config - My Printers: Scanner

The fastest way to add a printer is to let PrintDeck identify compatible devices on your current local network.



## FIND MY PRINTER ON NETWORK

### How the scanner works

Open My Printers, choose Add printer, then select Find my printer on network. PrintDeck immediately shows results found during the last five minutes while starting a fresh scan of the current local subnet. New matches appear under Found in this scan; cached matches remain under Recently found with their age. A scan never adds, edits or selects a printer by itself.

## AFTER A MATCH

### Verify and add several printers

Choose Add beside a result. Moonraker identity and access are checked automatically; Bambu Lab asks for the LAN access code and verifies an authenticated local MQTT/TLS report. Only a successful Check & add saves the profile. PrintDeck then returns to the scanner, marks that result Added and keeps the other matches ready for the next printer.

# Web Config - Moonraker Printer

Connect Klipper-based printers directly through the Moonraker HTTP API available on your local network.

## CONNECTION FIELDS

### Address the Moonraker service

Printer name is the friendly label shown throughout PrintDeck. Enter the printer IP address or hostname and its Moonraker port; 7125 is the usual default. An API key is optional on a trusted LAN and is needed only when your Moonraker configuration requires one.

## PRINTER IDENTITY

### Detect, review and connect

Detect printer reads Moonraker information and printer.cfg without changing the printer. When recognized, it suggests the manufacturer and model; both fields remain editable. Check & add verifies the current address and API key before saving. When adding from a scan, PrintDeck returns to the result list so more printers can be added without losing the recent matches.

# Web Config - Bambu Lab Printer

Bambu Lab printers connect directly over the local MQTT/TLS service. No Bambu cloud account is used.

## LOCAL LAN CONNECTION

### Collect three values from the printer

Enter the printer IP address or hostname and its serial number. To find the LAN access code, open the LAN Only Mode or Developer Mode screen on the printer and copy the code shown there; neither option needs to be enabled. Check & add requires one authenticated MQTT/TLS report before the profile is saved. The code remains on PrintDeck and is never sent to an external service.

## OPTIONAL COMPATIBILITY REPORT

### Help improve PrintDeck

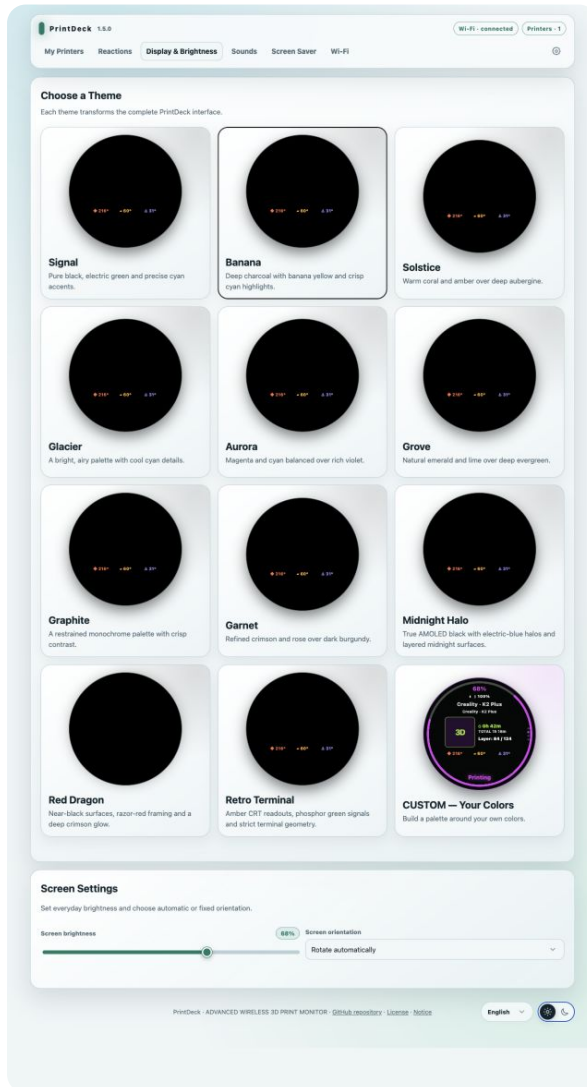
The report records the fields exposed by the printer and how their values are formatted. Model and firmware differences can introduce new fields or safe numeric ranges, so sharing it helps PrintDeck support more Bambu Lab printers. The report excludes connection details, access codes and print names. It stays on your device: nothing is uploaded or sent automatically. Download the report and email it to us only if you choose to share it.

Send the downloaded report to [support@printdeck.xyz](mailto:support@printdeck.xyz)

# Web Config - Display & Brightness

Shape the PrintDeck interface and see brightness, orientation and theme choices reflected on the device immediately.

## THEMES - COMPLETE SECTION



### THEMES

#### Choose a visual system

Theme cards preview printer data and status colours before you choose one. Custom lets you edit and save the complete palette.

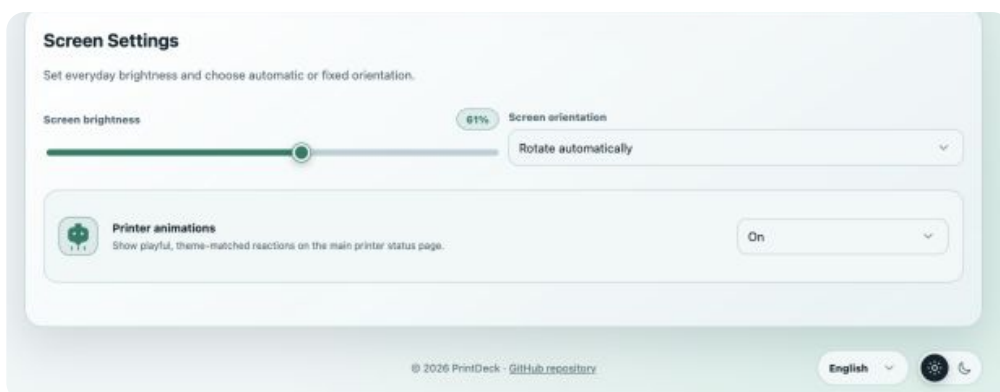
### SCREEN SETTINGS

#### Brightness & orientation

Set everyday brightness and choose automatic orientation or lock the screen to a fixed rotation.



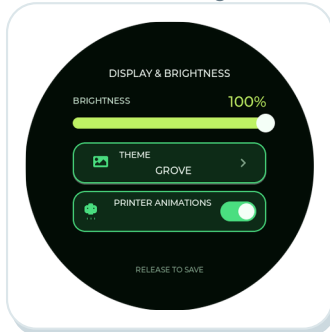
## SCREEN SETTINGS - BRIGHTNESS & ORIENTATION



# Reactions

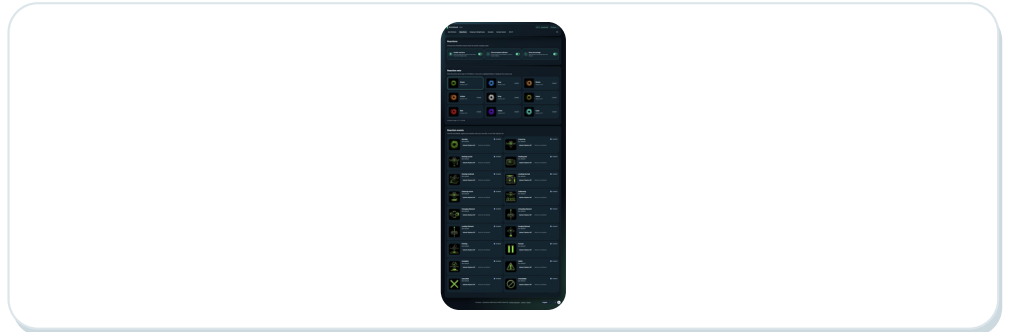


Reactions add a first printer page whose animation follows the current activity. Web Config has three independent visibility switches: Reactions, progress ring or bar, and percentage. The ring or bar and percentage are on by default. Round displays use a ring; square displays use a bar. Printer Status stays unchanged.



## ON PRINTDECK

Long-press a regular screen, open Display & Brightness and use the Reactions switch.



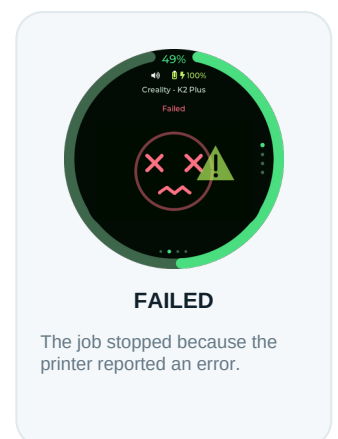
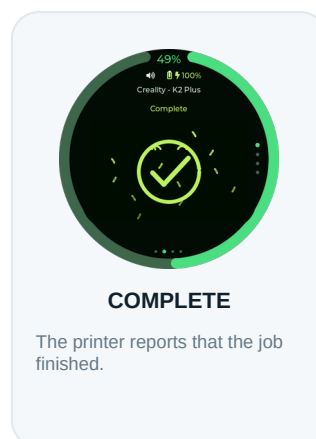
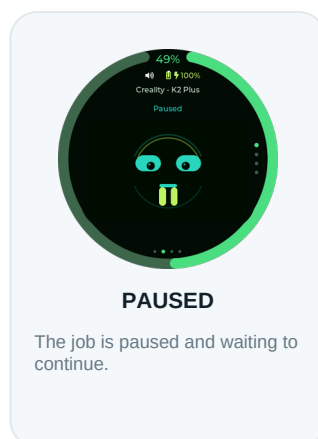
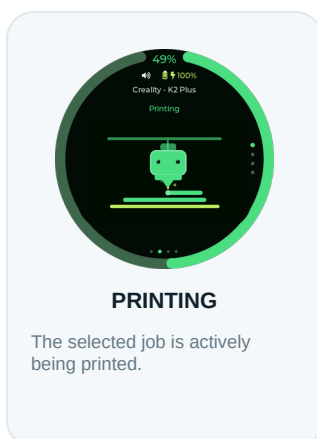
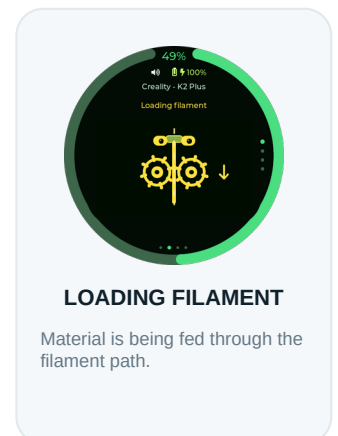
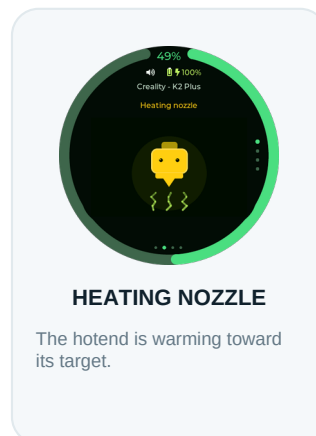
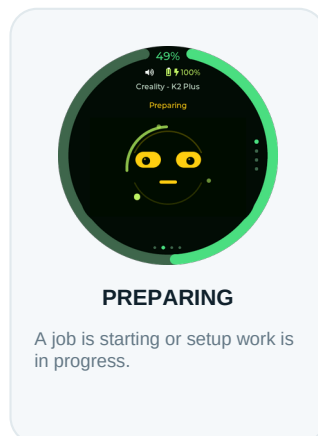
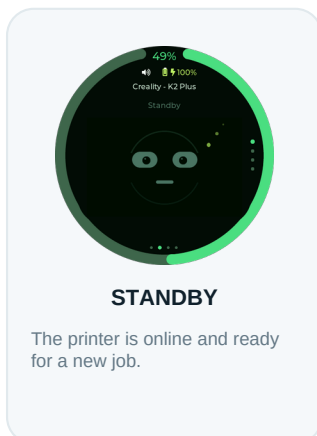
## IN WEB CONFIG

Open Reactions to control three switches independently: Reactions, progress ring or bar, and percentage. Both progress controls start on. Choose one of three visual sets, then preview, disable, replace or reset each supported printer event.



## WHAT EACH REACTION MEANS

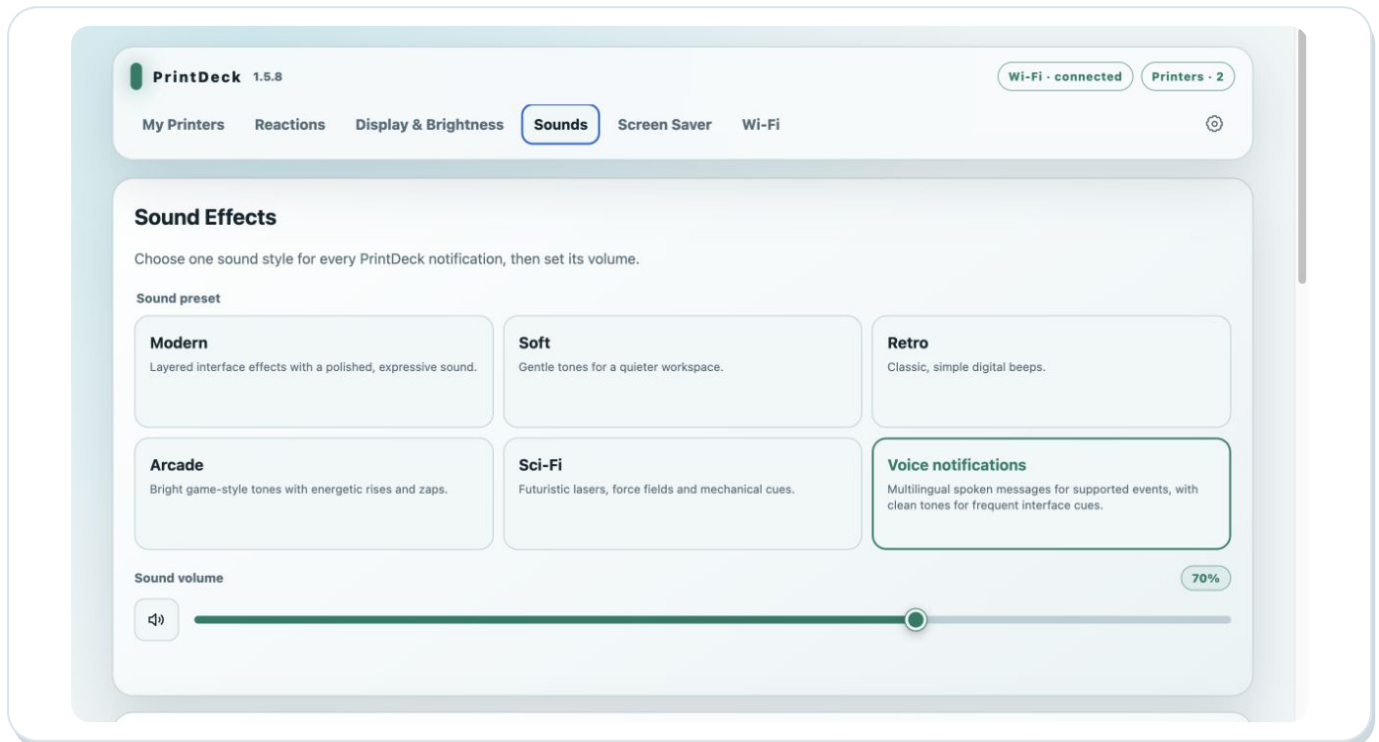
Captured from a physical round PrintDeck



Set downloads are validated before activation. Custom GIF overrides stay active across set changes and are removed only with Reset to set default.

# Web Config - Sounds

Choose how PrintDeck confirms actions and announces important printer events in your workspace.



## SOUND PRESET

### Six distinct sound sets

Choose Modern, Soft, Retro, Arcade, Sci-Fi or Voice Notifications. Voice Notifications speaks supported events in the selected language and uses clean tones for frequent interface cues.

## VOLUME & MUTE

### Match the room

Set the master volume or mute sound completely. Individual notification events can also be muted without disabling all feedback.

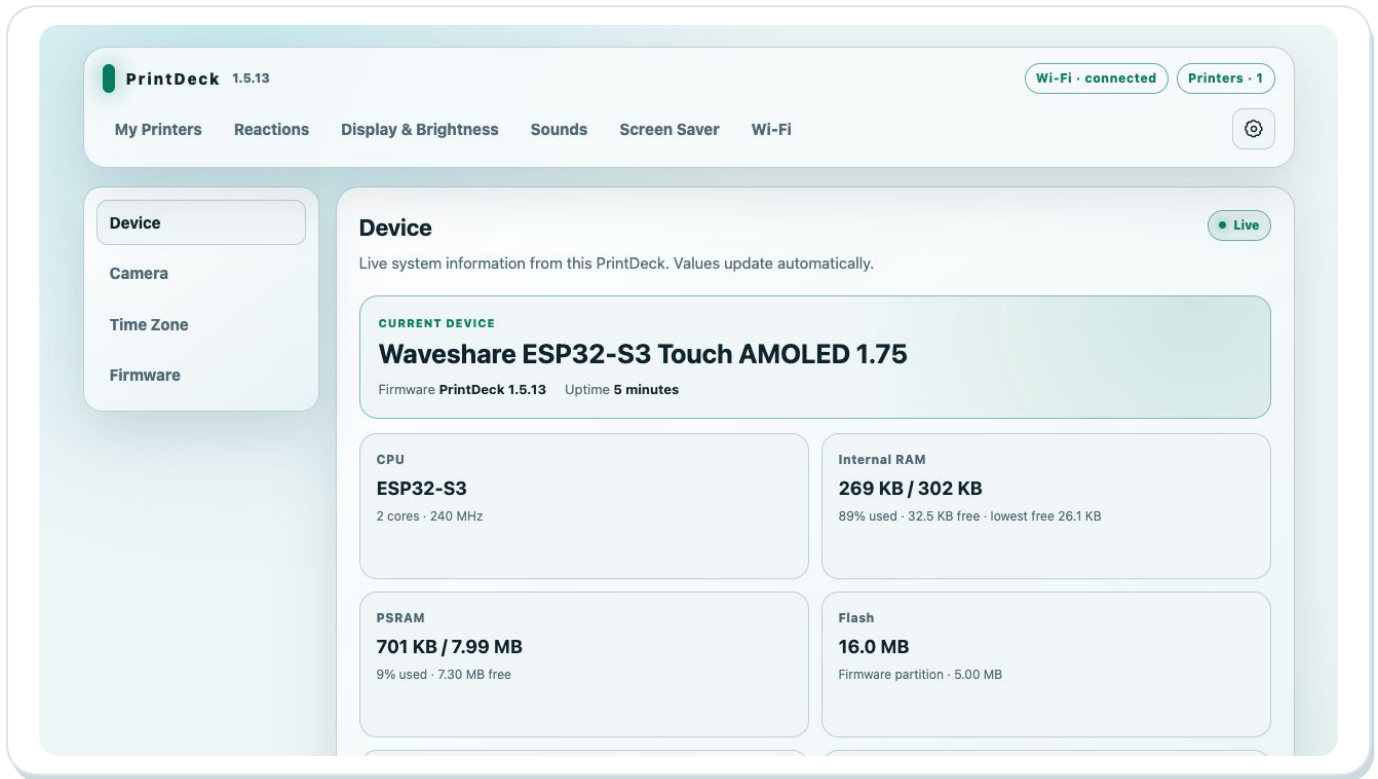
## TEST SOUND

### Preview before you finish

Use Test sound beside an event to hear the selected preset and volume. Testing plays a preview only and does not change saved event settings.

# Web Config - Device & System Settings

Review the current PrintDeck hardware state, then use the remaining tabs to manage idle behaviour, local connectivity, camera preferences, time zone and firmware.



## DEVICE

### Current state, never history

See the hardware model, firmware, uptime, CPU, current RAM and PSRAM use, Flash capacity, network signal, display and system version. Values refresh only while Device is open and are not stored.

## SCREEN & NETWORK

### Idle safely and stay local

Set dim and display-off timers, choose wake behaviour and connect PrintDeck to the same local Wi-Fi as your printers. Recovery mode remains available when a saved network cannot be reached.

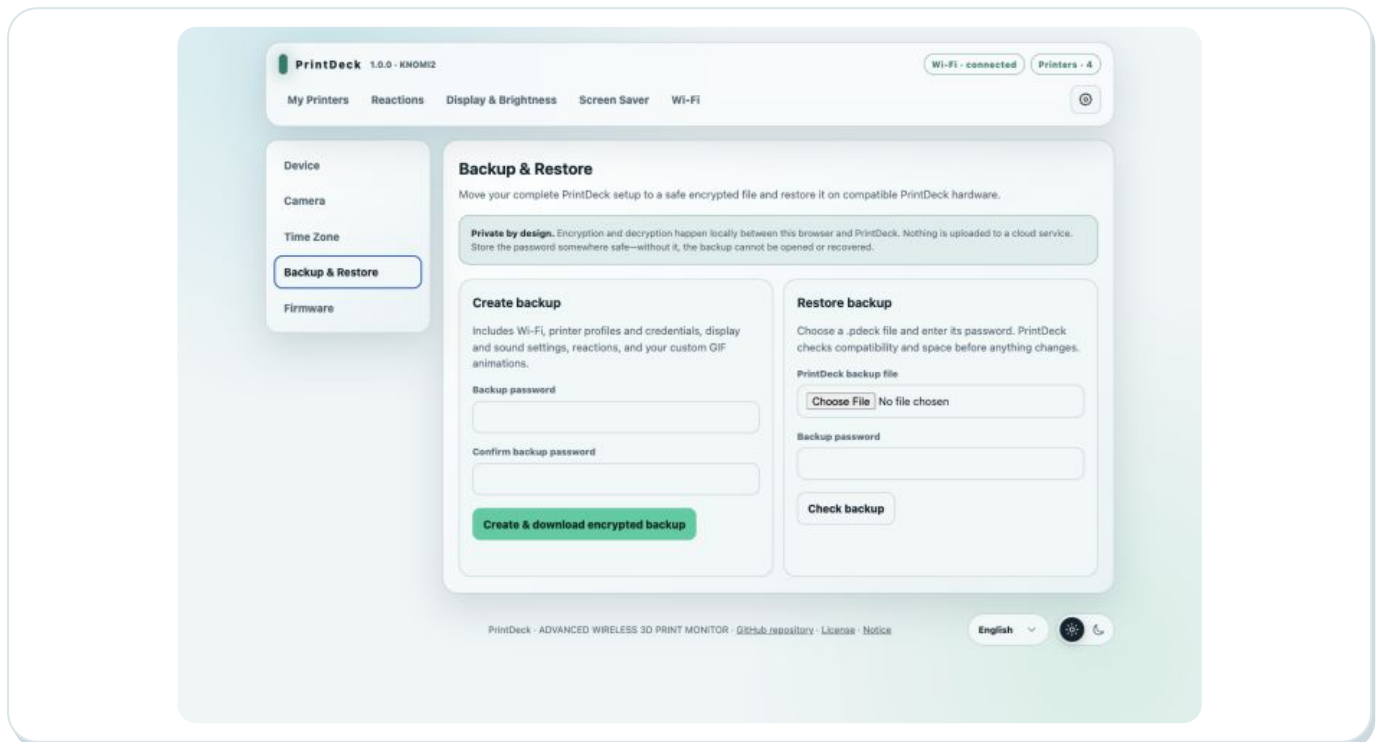
## CAMERA, TIME & UPDATES

### Tune the remaining device settings

Choose the camera refresh mode, set the time zone used by clocks and install validated PrintDeck firmware. Keep the device powered while an update is validated and restarted.

# Web Config - Backup & Restore

Save the complete PrintDeck setup before an update, reset or device replacement, then restore it on the same compatible PrintDeck hardware.



## PASSWORDS CANNOT BE RECOVERED

Keep the .pdeck file and its password somewhere safe. Without that exact password, neither PrintDeck nor support can open or restore the backup.

### CREATE A BACKUP

#### One encrypted .pdeck file

Open Device Settings, then Backup & Restore. Enter a password of at least 12 characters twice and choose Create & download encrypted backup. The file includes Wi-Fi and printer credentials, printer profiles, display and sound settings, reaction choices and every custom GIF animation currently stored on PrintDeck. Save the password separately from the file so losing one copy does not expose or strand the other.

### RESTORE SAFELY

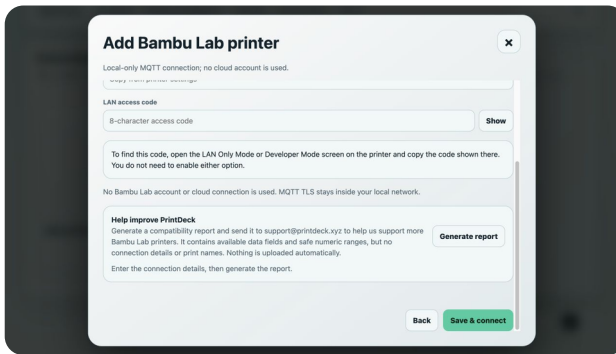
#### Check first, change second

Choose the .pdeck file, enter its password and select Check backup. Review the hardware, printer count, custom-animation count and file size before restoring. Keep current Wi-Fi settings selected when using your normal network so the page can reconnect after restart. PrintDeck restores custom GIFs one at a time, validates the same per-file and total storage limits, saves configuration last and then restarts.

Credentials, settings and custom GIF content are protected by AES-256-GCM. The key is derived locally from the password; only compatibility and size metadata remain readable. Nothing is uploaded.

Whether you are checking a new printer, choosing compatible third-party components or building your own PrintDeck display, support starts with a direct conversation.

#### HELP IMPROVE PRINTDECK



#### DOWNLOAD THE REPORT

##### Generate locally, then share it yourself

In Web Config open My Printers, choose Add printer, select Bambu Lab and fill in the local connection details. Choose Generate report, review the result and download it. PrintDeck never uploads or sends it automatically. Email the downloaded file to us only when you decide to share it.

#### LAN-ONLY PRIVACY

Printer credentials and compatibility data remain on PrintDeck and your local network. The report excludes access codes and other connection secrets; you control if and when it leaves your LAN.

#### COMPONENT GUIDANCE

##### Planning your own build?

Ask for help choosing a compatible third-party display, cables and optional parts for your own PrintDeck build. PrintDeck is firmware and does not offer a branded ready-to-use device.

#### INDIVIDUAL ASSEMBLY HELP

##### Need help putting it together?

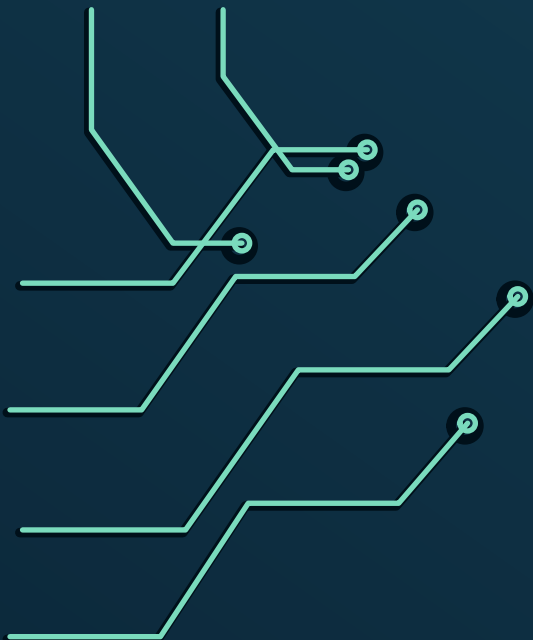
Ask about individual help assembling and configuring components for your personal build. Parts remain third-party products and are documented separately from the assembly service; no boxed PrintDeck hardware set is offered.

#### QUESTIONS & COLLABORATION

##### Tell us what you want to build

For compatibility reports, component guidance, individual assembly help, general questions or collaboration, write to [support@printdeck.xyz](mailto:support@printdeck.xyz). Include your printer model and the kind of help you need so we can respond with the right next step.

[support@printdeck.xyz](mailto:support@printdeck.xyz)



# PrintDeck

## PRINT SMARTER. STAY PRESENT.

A local-first printer companion for the displays you can keep exactly where you need them.



START HERE

### Manual, installer and case designs

Scan to open the PrintDeck website, then choose the path that matches your device and build.

<https://printdeck.xyz/>  
[support@printdeck.xyz](mailto:support@printdeck.xyz)



AMOLED + LCD

LOCAL BY DESIGN